

Topics:

1. n linear equations, n unknowns
2. Row picture
3. column picture
4. Matrix Form

2 equations 2 unknowns

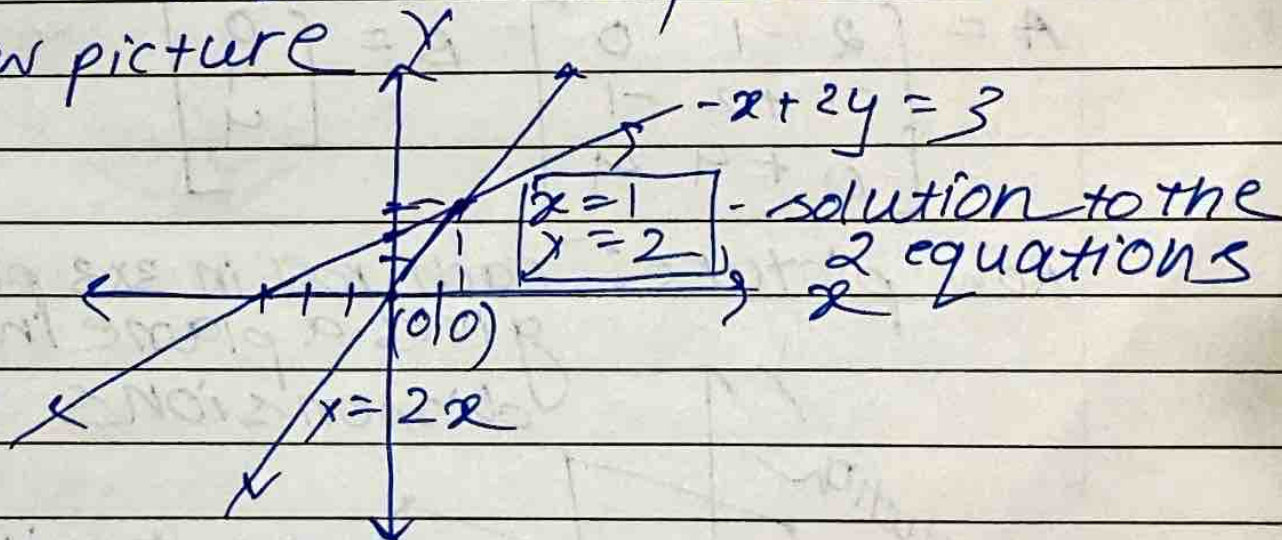
$$2x - y = 0$$

$$-x + 2y = 3$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

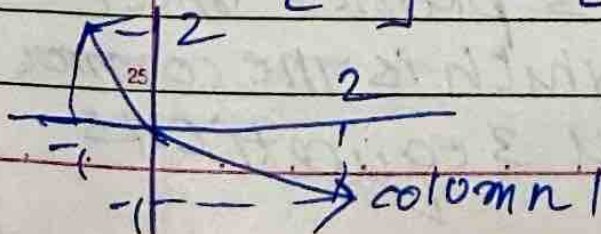
$AX = b$ - vectors/matrices

Row picture

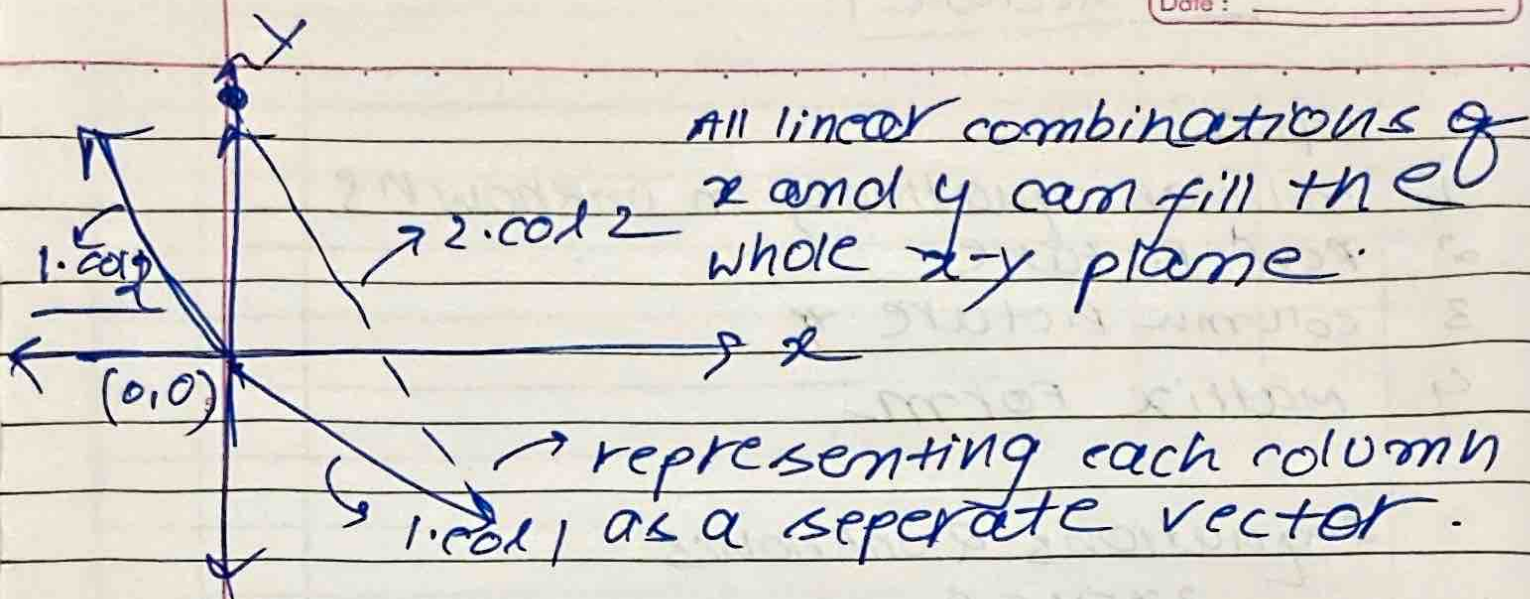


column picture

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \text{ - linear combination of columns}$$



put $x=1, y=2$ as trial combination



3 equations 3 unknowns

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$$2x - y = 0$$

$$-x + 2y + z = -1$$

$$-3y + 4z = 4$$

$AX = B$ - matrix form

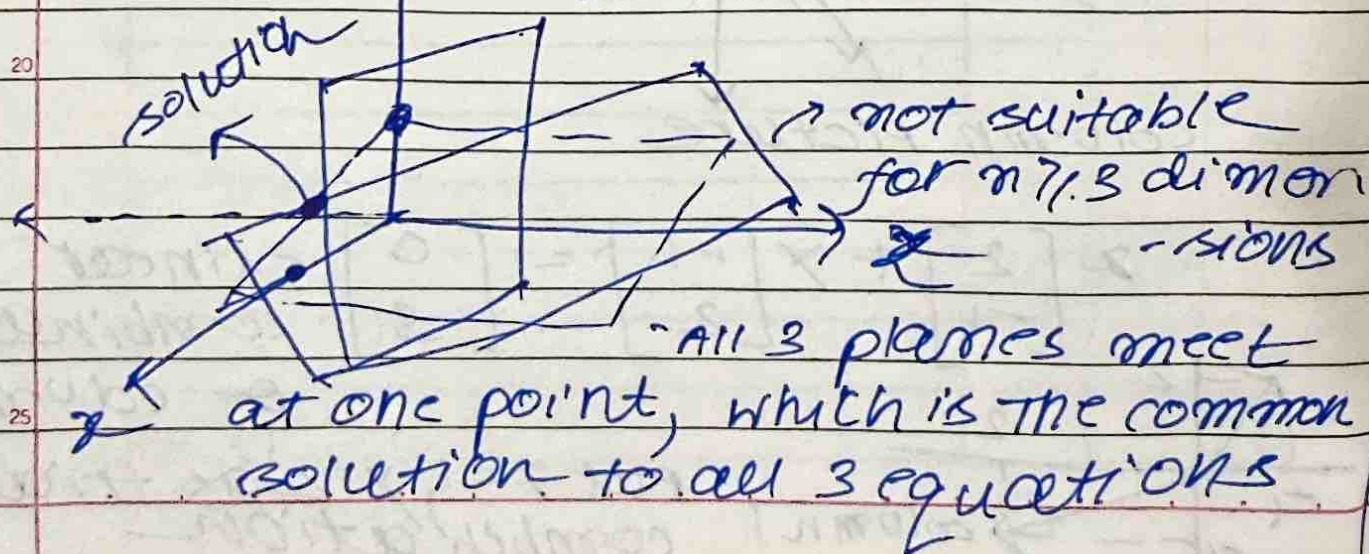
- will be solved in 3-dimensions

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$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & +2 & -1 \\ 0 & +4 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

Row picture: Each row in 3×3 problem gives a plane in 3-dimensions

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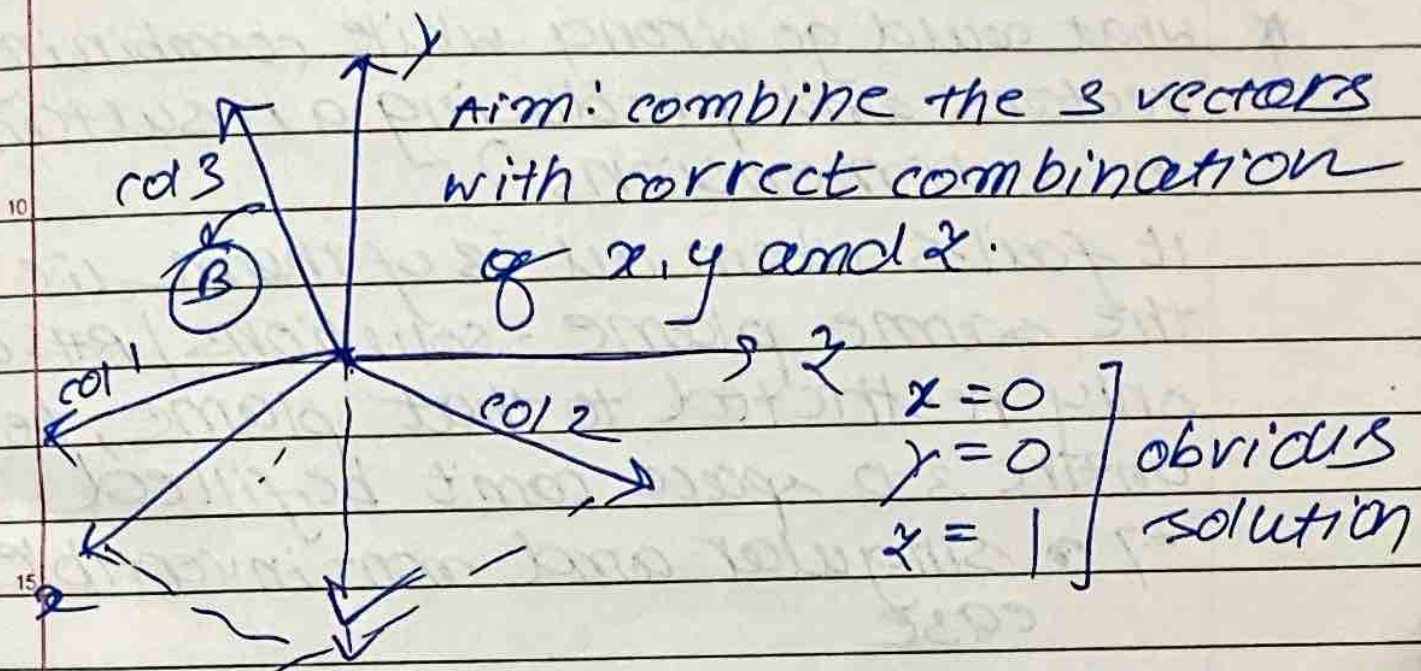


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column picture:

$$x \begin{bmatrix} +2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

LHS: linear combination of 3 vectors
each line is a vector in 3 dimensions.



large-scale production algorithms use this method to find all possible values of variables with all possible values in the RHS.

Row picture only gave solution to one possible RHS. Column picture can be used to combine any values of variables to produce any possible numeric result.

Q can we solve $Ax=b$ for every b ?
 Q do the linear combinations of the columns fill the entire 3-D space?

5 ~~yes~~ yes, for non-singular ($|A| \neq 0$) and invertible matrices.

10 * what could go wrong while combining 3 vectors and producing a resultant in 3-dimensions?

it fails when all 3 of them lie in the same plane. solutions/RHS are only restricted to that plane, hence entire 3-D space can't be filled.

15 $\Rightarrow$ singular and non-invertible case

20 Think about 9 equations & 9 unknowns we will have to do linear combinations of 9 vectors in 9-dimensional space $\Rightarrow$ 9 components, 9 rows/columns

it will yield good results.

if any $[a_r] = [a_{RH}]$, $0 < r < 9$, $r \in I$

$[]$ -column - then it will fail

25 here we will be restricted several 8-dimensional planes in a 9-D space

Multiplying a matrix by a vector

method 1: eg: $Ax \equiv b$ - column method

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}$$

method 2:- dot product

Ax - a linear combination of columns of A .